



A Decade of UPI

India's Digital Payment Revolution — Finance, Banking and Economic Impact

A RESEARCH REPORT

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Executive Summary

The Unified Payments Interface (UPI) is India's flagship real-time digital payment system, developed by the National Payments Corporation of India (NPCI) and launched in 2016 to create a simple, compatible, and instant payment ecosystem. Built on top of the Immediate Payment Service (IMPS), UPI transformed the way individuals and businesses transfer money by enabling seamless bank-to-bank transactions through mobile devices without requiring traditional banking details. Over the past decade, it has evolved from a digital payments initiative into one of India's most significant Digital Public Infrastructure (DPI) projects and is now widely regarded as a global benchmark for real-time retail payments.

This report commemorates the completion of ten years of UPI by documenting its evolution from a nascent payment interface into one of the world's largest digital payment networks. It examines how technological innovation, supportive public policy, financial inclusion initiatives, and collaboration among regulators, banks, and fintech companies have transformed India's payment landscape. The report also explores UPI's broader economic impact, its influence on banking and commerce, and the opportunities and challenges that lie ahead.

Several milestones have shaped UPI's decade-long journey. Following its pilot launch on 11 April 2016, applications such as PhonePe, BHIM, and later Google Pay accelerated adoption, while demonetisation in late 2016 further boosted digital payments. Subsequent innovations, including UPI 2.0, QR-based merchant payments, UPI Lite, RuPay Credit Cards on UPI, international payment linkages, and ATM cash withdrawals through UPI, expanded its use beyond peer-to-peer transfers. By 2025, UPI had crossed 20 billion monthly transactions, becoming the backbone of India's retail payments ecosystem.

According to the Press Information Bureau's ten-year UPI release, the platform processed over 24,000 crore transactions during FY 2024–25, with a total transaction value exceeding INR 314 lakh crore. More than 460 million unique users, 65 million merchants, and over 675 banks now participate in the ecosystem. India accounts for nearly half of all global real-time digital payment transactions, with UPI driving this leadership position and demonstrating the success of India's Digital Public Infrastructure model in promoting financial inclusion, reducing transaction costs, and supporting the formalisation of the economy.

As UPI enters its second decade, sustaining this growth will require addressing challenges related to cybersecurity, fraud prevention, infrastructure scalability, merchant monetisation, and regulation. Beyond being a payment platform, UPI is expected to continue evolving as a foundational layer of India's digital economy.

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Chapter 1: Introduction

1.1 India's Payment Landscape Before UPI

Before the introduction of UPI in 2016, India's digital payments ecosystem was fragmented, complex and heavily dependent on traditional banking infrastructure. Although electronic payment systems such as the National Electronic Funds Transfer (NEFT), Real-Time Gross Settlement (RTGS), and Immediate Payment Service (IMPS) had enabled digital fund transfers, they were designed primarily for banking operations rather than everyday consumer transactions. These systems often required users to enter lengthy bank account details, IFSC codes and beneficiary information, making routine payments cumbersome and less accessible for the average consumer.

At the same time, India's economy remained overwhelmingly cash-driven. According to the Reserve Bank of India (RBI), cash continued to dominate retail transactions, particularly among small merchants, informal businesses and rural communities, where digital payment acceptance remained limited. Although the rapid growth of smartphones and mobile internet presented new opportunities for digital commerce, the absence of a unified, compatible payment ecosystem meant that consumers often relied on closed-loop mobile wallets or cash for everyday transactions.

This fragmented environment highlighted several structural challenges. Payment systems lacked interoperability across banks and service providers, merchant acceptance infrastructure remained underdeveloped, and complex user experiences constrained digital payment adoption. These limitations also hindered broader objectives such as financial inclusion, digital commerce and the formalisation of India's economy. Consequently, there emerged a need for a payment infrastructure that could facilitate instant, secure and seamless bank-to-bank transfers while remaining accessible to users across all segments of society.

1.2 Birth of the Unified Payments Interface

Recognising these challenges, the National Payments Corporation of India (NPCI), under the guidance of the Reserve Bank of India, developed UPI as a next-generation retail payment system. Officially launched with a pilot involving 21 banks on 11 April 2016, UPI was built upon the IMPS infrastructure but introduced a fundamentally different approach to digital payments by enabling compatible, real-time bank-to-bank transactions through a single mobile application.

Unlike existing payment systems, UPI eliminated the need for users to repeatedly enter bank account details or beneficiary information. Instead, transactions could be initiated using a Virtual Payment Address (VPA), mobile number, QR code or other simplified identifiers, significantly improving the speed and convenience of digital payments. Its open architecture also enabled banks, fintech companies and third-party application providers to innovate on a

common payments infrastructure, fostering competition while maintaining compatibility across the ecosystem.

The launch of UPI represented more than a technological advancement; it marked a paradigm shift in India's payment infrastructure. By combining instant settlement, compatibility and a mobile-first design, UPI laid the foundation for a scalable digital payments ecosystem capable of serving hundreds of millions of users.

1.3 Objectives of UPI

The primary objective of UPI was to simplify digital payments while creating a compatible ecosystem that could support India's growing digital economy. Designed as a public digital infrastructure, UPI sought to make electronic payments as convenient as exchanging cash, irrespective of the user's bank or payment application.

Its key objectives included:

- Enabling instant, 24×7 bank-to-bank fund transfers.
- Promoting compatibility across banks and payment service providers.
- Supporting financial inclusion through simple and accessible payment mechanisms.
- Reducing reliance on cash for retail transactions.
- Lowering transaction costs for consumers and merchants.
- Encouraging innovation within India's fintech ecosystem through open APIs and standardised infrastructure.
- Strengthening the transparency and efficiency of the country's payment system.

Collectively, these objectives aligned with the Government of India's broader vision of creating a digitally empowered economy supported by robust public digital infrastructure.

1.4 Scope and Objectives of this Report

This report presents a comprehensive assessment of the Unified Payments Interface's first decade of operation, examining its evolution from a domestic payment innovation into one of the world's largest real-time payment systems. It analyses the technological, regulatory and economic developments that have shaped UPI's growth, while evaluating its contribution to financial inclusion, merchant digitisation, banking transformation and India's broader digital economy.

Drawing upon primary sources including the National Payments Corporation of India, the Reserve Bank of India and the Press Information Bureau (PIB), alongside insights from leading consulting firms, industry publications and financial newspapers, the report combines quantitative analysis with qualitative perspectives to provide a balanced view of UPI's achievements, challenges and future trajectory. Through this analysis, the report aims to understand not only how UPI transformed India's payment ecosystem over the past decade,

but also how it is positioned to influence the future of digital payments both domestically and globally.

Chapter 2: The Evolution of UPI (2016–2026)

The evolution of the UPI over the past decade represents one of the most significant transformations in India's financial services sector. What began as a connected payment platform developed by NPCI has evolved into the backbone of India's digital payments ecosystem, processing over **24,162 crore annual transactions** valued at approximately INR 314 lakh crore during FY 2025–26. Today, UPI accounts for nearly 85% of India's digital payment transactions and almost 49% of all global real-time payment transactions, demonstrating its scale and international significance.

Rather than experiencing linear growth, UPI's journey has unfolded across four distinct phases: Foundation (2016–2017), Mass Adoption (2018–2020), Hyper Growth (2020–2022), and Global Expansion & Innovation (2023–2026). Each phase introduced technological advancements, regulatory support, and ecosystem innovations that collectively transformed UPI from a payment interface into one of the world's most successful examples of Digital Public Infrastructure.

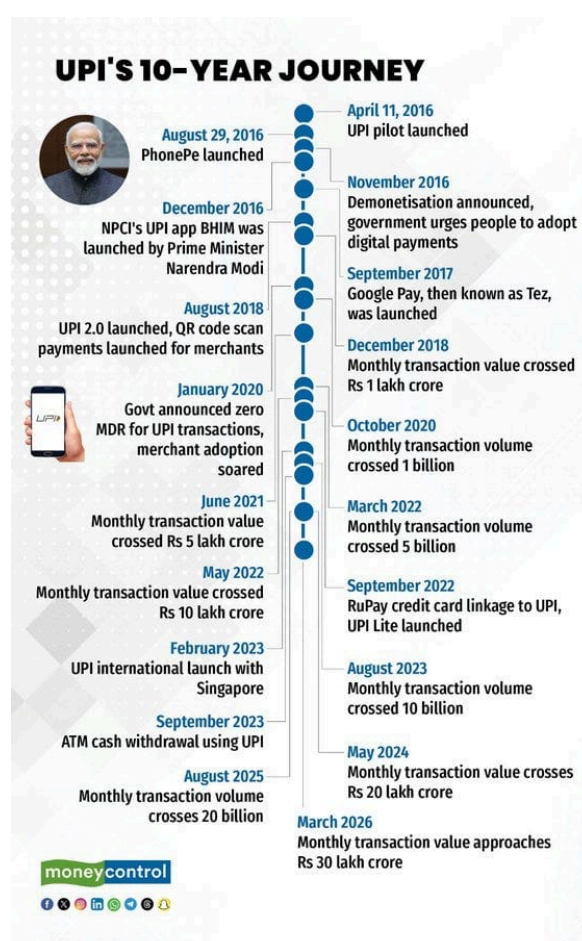


Figure 1: Key milestones in UPI's decade-long growth, highlighting the rapid expansion of digital payments in India

2.1 Phase I – Foundation (2016–2017)

The foundation of UPI was laid on 11 April 2016, when the National Payments Corporation of India, under the regulatory oversight of the Reserve Bank of India, launched a pilot involving 21 banks. Built upon the Immediate Payment Service, UPI introduced a fundamentally new payments architecture by enabling real-time, cross-bank transfers through a mobile-first platform. Unlike existing payment systems, users could send or receive money using a Virtual Payment Address (VPA), mobile number, or QR code, eliminating the cumbersome process of repeatedly entering bank account details and IFSC codes.

The ecosystem gained early momentum with the launch of PhonePe in August 2016 and the government's BHIM (Bharat Interface for Money) application in December 2016. The announcement of demonetisation in November 2016 further accelerated digital payment adoption, encouraging consumers and merchants to shift toward cashless transactions. While transaction volumes remained modest during the initial months, this period established the technological and regulatory foundation that would support UPI's subsequent expansion.

2.2 Phase II – Mass Adoption (2018–2020)

Between 2018 and 2020, UPI transitioned from an emerging payments platform to a mainstream consumer payment system. The introduction of UPI 2.0 enhanced functionality through features such as overdraft account support, signed payment mandates, invoice verification, and improved merchant payment capabilities. Simultaneously, the rapid deployment of compatible QR codes significantly simplified merchant acceptance, allowing businesses of all sizes to participate in the digital payments ecosystem.

Competition among fintech platforms, including PhonePe, Google Pay (formerly Tez), and Paytm, improved user experience and accelerated consumer adoption. By December 2018, monthly transaction values had crossed INR 1 lakh crore, while continued innovation and widespread smartphone penetration drove sustained growth. In January 2020, the Government of India removed Merchant Discount Rate (MDR) charges on UPI transactions, reducing payment acceptance costs for merchants and further encouraging digital payment adoption across the country.

2.3 Phase III – Hyper Growth (2020–2022)

The COVID-19 pandemic marked a turning point in UPI's adoption. As consumers and businesses increasingly preferred contactless payment methods, UPI experienced unprecedented growth in both transaction volume and value. Digital payments became an essential component of everyday economic activity, spanning retail purchases, utility payments, peer-to-peer transfers, and small business transactions.

The scale of adoption during this period was remarkable. Monthly transaction volume crossed one billion in October 2020, exceeded five billion by March 2022, and monthly

transaction values surpassed INR 10 lakh crore shortly thereafter. Continuous innovation also expanded the platform's capabilities through the introduction of UPI Lite for low-value transactions and the integration of RuPay Credit Cards on UPI, enabling credit-based payments directly through the UPI ecosystem. These developments reinforced UPI's position as the dominant retail payment system in India.

2.4 Phase IV – Global Expansion and Innovation (2023–2026)

The most recent phase has been characterised by internationalisation and ecosystem diversification. Beginning with the launch of cross-border payment connectivity with Singapore in 2023, UPI expanded its global presence through partnerships with multiple countries, reflecting growing international recognition of India's Digital Public Infrastructure model. The ecosystem also introduced innovations such as UPI-based ATM cash withdrawals, enhanced credit products, and broader merchant services, extending UPI beyond conventional peer-to-peer payments.

By FY 2025–26, UPI had evolved into the world's largest real-time retail payment system. Annual transaction volume exceeded 24,162 crore transactions, with transaction value reaching approximately INR 314 lakh crore. More than 700 banks had joined the network, while UPI accounted for approximately 85% of India's digital payments and nearly 49% of global real-time payment transaction volume. These achievements illustrate how continuous innovation, strong regulatory support, and widespread ecosystem participation transformed UPI from a domestic payment solution into a globally recognised model for digital financial infrastructure.

2.5 End-of-Chapter Key Insights

- ❖ UPI's evolution can be understood through four distinct phases: Foundation, Mass Adoption, Hyper Growth, and Global Expansion.
- ❖ Government policy, regulatory support, fintech innovation, and smartphone penetration collectively accelerated adoption.
- ❖ Continuous product innovation, including QR payments, UPI 2.0, UPI Lite, and credit integration, expanded the platform's use cases beyond simple fund transfers.
- ❖ Within a decade, UPI became the world's largest real-time payment system and a flagship example of India's Digital Public Infrastructure.

Chapter 3: UPI by the Numbers

The rapid expansion of the Unified Payments Interface over the past decade is best understood through its quantitative growth. From processing only a few thousand transactions during its pilot phase in 2016, UPI has evolved into the world's largest real-time retail payment system, supporting billions of monthly transactions across consumers, merchants, businesses and government entities. The platform's remarkable growth reflects not only increasing digital adoption but also the success of India's Digital Public Infrastructure model in creating an interoperable, scalable and inclusive payment ecosystem.

According to the Press Information Bureau, UPI processed 24,162 crore transactions during FY 2024–25, representing a total transaction value exceeding INR 314 lakh crore. Today, over 460 million users, 65 million merchants, and 675+ participating banks form part of the UPI ecosystem, while India contributes nearly 49% of global real-time digital payment transactions. These metrics illustrate the scale at which UPI has transformed India's payment landscape and established itself as a global benchmark for digital payments.

3.1 Growth in Transaction Volume

Transaction volume has been the most visible indicator of UPI's success. During its initial years, adoption was gradual as consumers and merchants familiarised themselves with digital payment applications. However, the combined effects of smartphone penetration, affordable mobile internet, fintech innovation and supportive government policies resulted in exponential growth from 2018 onwards. The COVID-19 pandemic further accelerated this trajectory as consumers increasingly shifted towards contactless and digital payment methods.

The platform reached several significant milestones during its first decade, including crossing one billion monthly transactions in 2019, five billion in 2022, and eventually exceeding twenty billion monthly transactions by 2025. This sustained growth demonstrates that UPI has evolved beyond a peer-to-peer payment platform into the default payment mechanism for retail transactions across India.

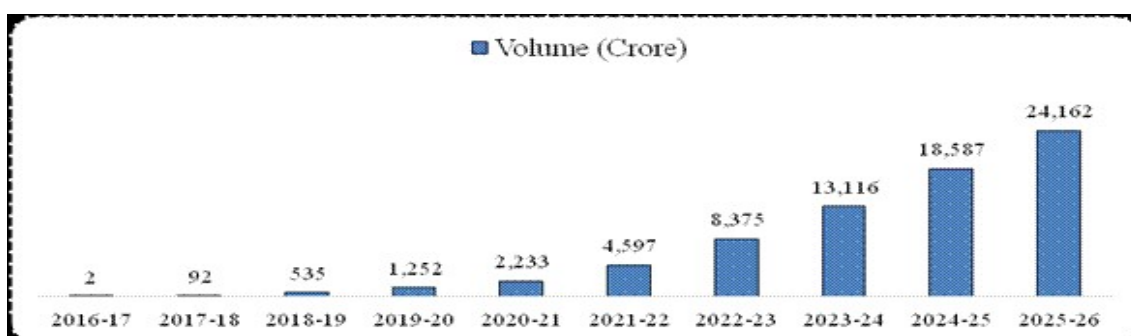


Figure 2: Exponential growth in UPI transaction volumes, reflecting the rapid adoption of digital payments and UPI's emergence as India's dominant retail payment platform

3.2 Growth in Transaction Value

While transaction volume reflects adoption, transaction value provides insight into the increasing economic significance of the platform. Initially dominated by low-value peer-to-peer transfers, UPI has progressively expanded into merchant payments, utility bill settlements, e-commerce transactions and business payments.

During FY 2024–25, UPI facilitated transactions worth over **INR 314 lakh crore**, highlighting its growing importance within India's financial system. The increasing average transaction value also suggests rising consumer confidence and broader acceptance across both retail and commercial use cases.

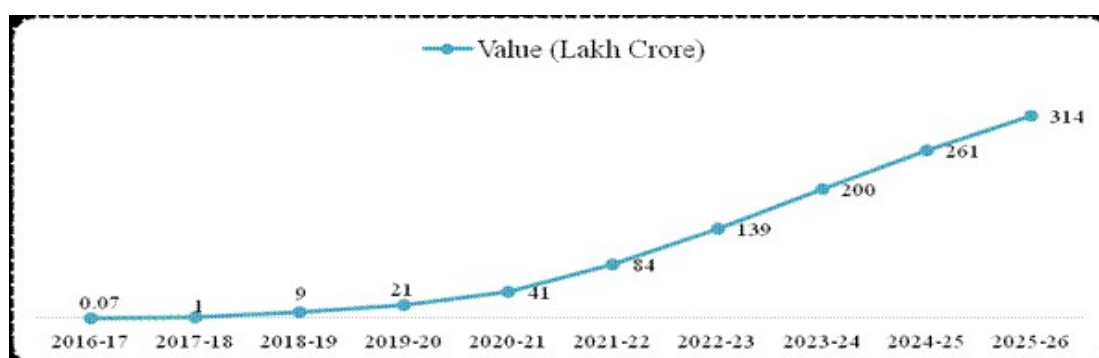


Figure 3: Rising UPI transaction value over the past decade, reflecting the expansion of UPI from low-value peer-to-peer transfers to a broader digital payments ecosystem

3.3 Expansion of the Banking Ecosystem

The success of UPI has been supported by the continuous expansion of participating financial institutions. What began with 21 banks during the pilot launch has grown into a network of over 703 banks, making UPI one of the world's most interoperable payment infrastructures.

This expansion reflects the collaborative nature of India's payment ecosystem, where public sector banks, private banks, small finance banks, payment banks and fintech companies operate on a common platform while maintaining healthy competition in customer-facing services.

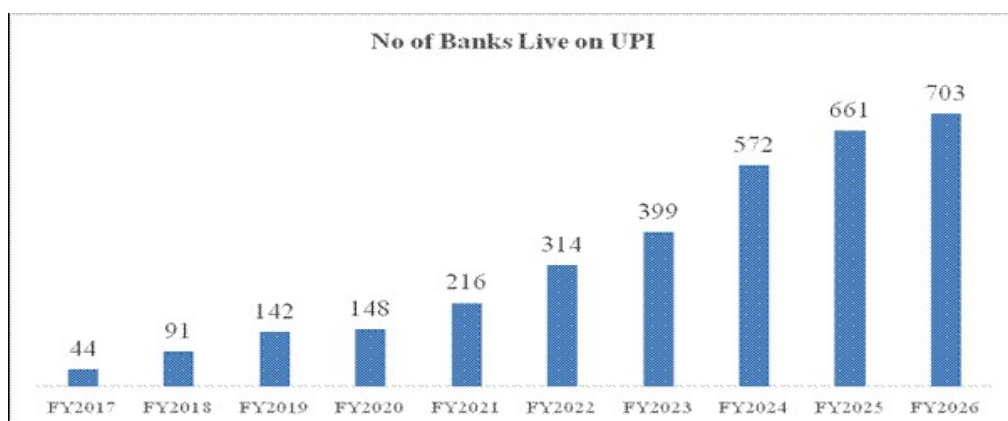


Figure 4: Expansion of UPI's participating-bank network, from its initial pilot to over 700 banks, reflecting the platform's growing reach and interoperability across India's financial ecosystem

3.4 Consumer and Merchant Adoption

UPI's widespread adoption is evident in the scale of its user base. More than 460 million unique users actively utilise the platform, while over 65 million merchants accept UPI payments across sectors ranging from organised retail to street vendors.

The rapid deployment of interoperable QR codes has significantly reduced the barriers to digital payment acceptance, enabling even micro-enterprises and small businesses to participate in the formal digital economy. This democratisation of payment infrastructure has played an important role in promoting financial inclusion and expanding digital commerce throughout the country.

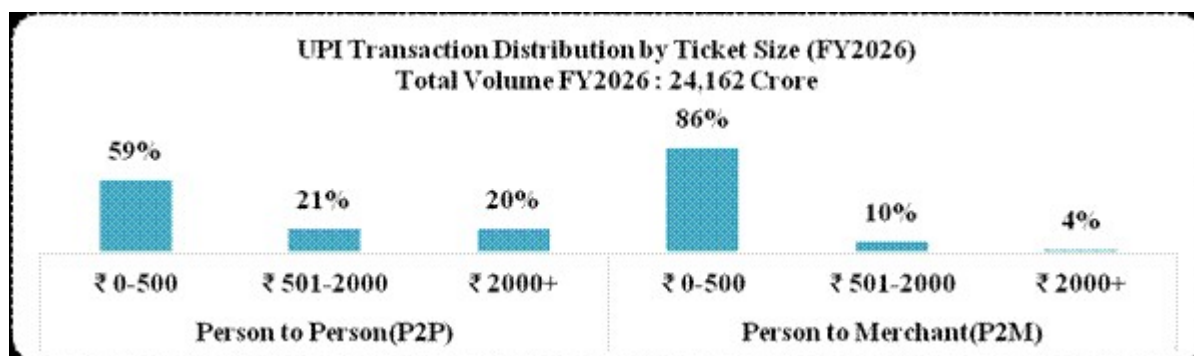


Figure 5: UPI transaction distribution by ticket size and transaction type in FY2026, highlighting the dominance of low-value payments across both P2P and P2M transactions.

3.5 India's Position in Global Real-Time Payments

UPI's success extends beyond India's domestic payment ecosystem. According to the Press Information Bureau, India now accounts for approximately 49% of global real-time payment transactions, making it the largest contributor worldwide.

The international recognition of UPI has led to growing interest from several countries seeking to replicate aspects of India's Digital Public Infrastructure model. Cross-border payment partnerships and international merchant acceptance continue to strengthen UPI's position as a global benchmark for interoperable digital payments.

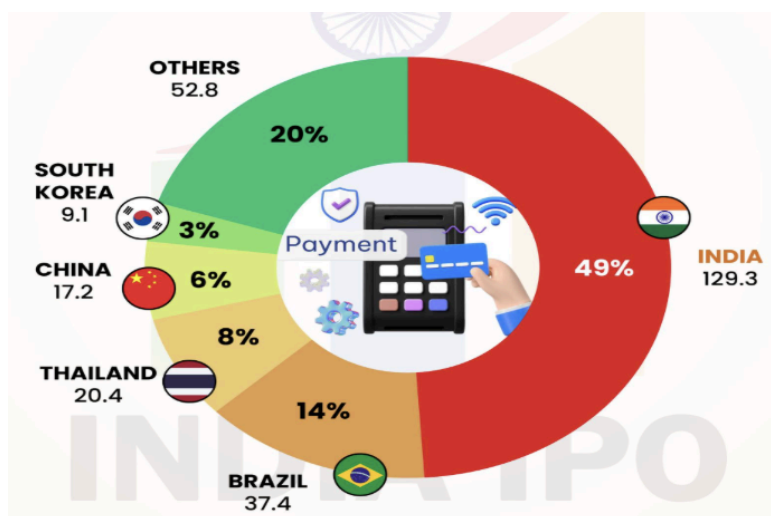


Figure 6: Global distribution of real-time payment transactions, with India accounting for the largest share and demonstrating the international scale and influence of UPI

3.6 Key Metrics at a Glance

Metric	FY 2025–26
Annual Transactions	24,162 Crore
Transaction Value	INR 314 Lakh Crore
Participating Banks	703
Unique Users	460 Million
Merchants	65 Million
Countries with UPI Presence	Multiple International Markets
Share of India's Digital Payments	85%
Share of Global RTP Transactions	49%

Chapter 4: Economic Impact of UPI

Over the past decade, the UPI has evolved beyond a digital payment mechanism to become a key driver in India's economic transformation. By enabling instant, low-cost and interconnected digital transactions, UPI has fundamentally altered the way individuals, businesses and governments exchange value. Its widespread adoption has accelerated financial inclusion, strengthened the formal economy, improved payment efficiency and fostered innovation within India's rapidly expanding fintech ecosystem.

4.1 Financial Inclusion

One of UPI's most significant contributions has been its role in advancing financial inclusion. Traditionally, access to formal financial services in India was constrained by geographical barriers, limited banking infrastructure and complex banking procedures. While initiatives such as the Pradhan Mantri Jan Dhan Yojana (PMJDY) expanded bank account ownership, many accounts remained underutilised due to the lack of accessible digital payment infrastructure.

UPI addressed this challenge by providing a simple, mobile-first payment platform that enabled individuals to transfer and receive money instantly using only a smartphone and a linked bank account. The interconnectivity of the system eliminated dependence on individual banking applications, allowing users from different banks to transact seamlessly. As smartphone penetration and internet connectivity improved across the country due to the launch of Reliance Jio, UPI significantly increased digital participation among previously underserved populations, particularly in semi-urban and rural regions.

Beyond enabling payments, UPI has acted as an entry point into the formal financial ecosystem. Regular digital transactions generate financial footprints that can support credit assessment, facilitate access to formal lending, and encourage greater participation in digital financial services. This has strengthened the broader objective of inclusive economic development.

4.2 Empowering MSMEs and Small Merchants

Micro, Small and Medium Enterprises (MSMEs) account for a substantial share of India's employment and economic output, yet historically faced challenges related to payment collection, cash management and limited access to formal credit.

UPI has significantly improved payment efficiency for these businesses by enabling instant settlements, reducing the dependence on cash and lowering transaction costs. The widespread adoption of QR codes has allowed even small vendors, street merchants and local retailers to accept digital payments without investing in expensive point-of-sale infrastructure.

The digital transaction records generated through UPI have also strengthened financial transparency, enabling businesses to maintain better accounting records and potentially improve their eligibility for formal credit products. This shift towards digital transactions has contributed to greater business formalisation while enhancing operational efficiency for millions of small enterprises.

4.3 Transforming Consumer Behaviour

UPI has fundamentally changed consumer payment habits by making digital transactions faster, more convenient and widely accessible. Payments that previously required cash, debit cards or lengthy bank transfer procedures can now be completed within seconds through mobile applications.

The platform's ease of use has encouraged digital payments across diverse use cases, including retail purchases, utility bills, transportation, food delivery, e-commerce, education and healthcare. Features such as QR-code payments, recurring payment mandates and integration with multiple banking applications have further improved the user experience.

As digital payments became embedded in everyday life, particularly following the COVID-19 pandemic, consumer trust in cashless transactions increased substantially. UPI has consequently played a central role in India's transition towards a digitally enabled consumption economy.

4.4 Impact on the Banking and Financial Services Sector

UPI has transformed the banking sector by shifting competition away from payment infrastructure towards customer experience and value-added financial services. Before UPI, banks largely controlled both payment processing and customer interaction. The introduction of this payment infrastructure enabled fintech companies to become the primary customer interface while banks continued to provide the underlying accounts and settlement mechanisms.

This model encouraged collaboration between traditional financial institutions and technology companies, leading to increased innovation across digital banking, lending, wealth management and embedded finance. Banks also benefited from lower cash handling costs, improved transaction efficiency and greater customer engagement through digital channels.

More broadly, UPI has strengthened India's fintech ecosystem by lowering barriers to innovation and enabling startups to develop new financial products using a common payments infrastructure.

4.5 Supporting Government Initiatives and the Formal Economy

UPI has complemented several government initiatives aimed at promoting transparency, financial inclusion and digital governance. By facilitating digital transactions, the platform has supported the broader objectives of Digital India while encouraging greater participation in the formal economy.

The widespread use of digital payments creates auditable transaction records, improving transparency and reducing dependence on cash-based transactions. This has enhanced efficiency in both public and private sector payments while supporting greater accountability within the financial system.

UPI has also strengthened the effectiveness of India's Digital Public Infrastructure by integrating with complementary initiatives such as Aadhaar, DigiLocker, Account Aggregator and Open Network for Digital Commerce (ONDC), collectively creating a more connected and digitally enabled economic ecosystem.

4.6 Catalysing India's Fintech Revolution

Perhaps the most enduring economic impact of UPI has been its contribution to India's fintech ecosystem. By providing open, interoperable payment infrastructure, UPI significantly lowered the barriers to entry for financial technology companies. Instead of building proprietary payment networks, startups could innovate on top of a common public infrastructure, directing resources toward improving customer experience and developing new financial products.

This ecosystem has facilitated the emergence of digital payment applications, embedded finance solutions, merchant platforms, lending technologies and financial management tools. The success of companies such as PhonePe, Google Pay, Paytm and numerous fintech startups demonstrates how public digital infrastructure can stimulate private sector innovation while maintaining competition and interoperability.

Increasingly, UPI is viewed internationally as a model for digital public infrastructure that balances public investment with private innovation, offering valuable lessons for other economies pursuing digital transformation.

Chapter 5: UPI and India's Financial Ecosystem

5.1 UPI as the Foundation of India's Digital Payments

The introduction of UPI fundamentally changed the role of banks within India's payment ecosystem. Traditionally, banks operated as both custodians of customer deposits and providers of payment infrastructure through services such as NEFT, RTGS, and IMPS. UPI shifted this paradigm by separating the payment interface from the banking infrastructure. While banks continue to hold customer accounts and process settlements, third-party applications such as PhonePe, Google Pay, and Paytm now manage much of the customer-facing payment experience.

This shift has encouraged banks to move beyond competing solely on payment channels and instead focus on developing stronger digital banking services, API-driven integrations, and value-added financial products. The result has been greater collaboration between banks and fintech companies, fostering an ecosystem where innovation is driven by competition in customer experience rather than proprietary payment networks.

5.2 Strengthening Public Finance and Government Services

UPI has enhanced the efficiency of government payment systems by enabling seamless digital transactions for taxes, utility bills, educational fees, transport services, and municipal payments. It complements broader government initiatives such as Direct Benefit Transfers (DBT), ensuring faster and more transparent movement of funds between public institutions and citizens.

From a public finance perspective, increased digital transactions contribute to improved audit trails and reduced cash handling costs. Greater transparency also supports tax compliance by creating digital records of commercial transactions, thereby strengthening revenue collection and reducing opportunities for leakages within the financial system.

5.3 Positioning India as a Leader in Digital Public Infrastructure

UPI has become one of the foundational components of India's Digital Public Infrastructure (DPI), alongside Aadhaar and DigiLocker. Together, these systems create an integrated ecosystem that enables secure digital identity verification, paperless documentation, and instant financial transactions.

India's DPI model has gained international recognition as a scalable and cost-effective framework for accelerating digital inclusion. Several countries are exploring partnerships with India to develop interoperable payment systems inspired by UPI, reflecting its growing influence as a global benchmark for public digital infrastructure. This international interest

underscores India's emergence as a leader in building technology-driven public financial systems.

5.4 Macroeconomic Impact

Beyond transforming individual payment experiences, UPI has generated broader economic benefits. The widespread shift toward digital transactions has reduced dependence on cash, lowering the costs associated with printing, transporting, and managing physical currency. Faster payment settlements improve the velocity of money within the economy, enabling businesses to receive funds instantly and enhancing overall economic efficiency.

Moreover, the vast volume of digital transaction data has created opportunities for improved credit assessment, financial analytics, and policy planning. For financial institutions, this data supports more accurate lending decisions, while for policymakers, it provides valuable insights into consumption patterns and economic activity. Collectively, these developments have strengthened financial inclusion, encouraged innovation, and contributed to the formalisation of India's economy.

Chapter 6: Challenges and Risks of UPI

After discussing UPI's transformative impact on India's financial ecosystem, it is equally important to examine the challenges that accompany its rapid growth. Despite its widespread adoption and global recognition, UPI faces issues related to cybersecurity, infrastructure scalability, market competition, profitability, and regulatory oversight. Addressing these challenges will be essential to ensuring the platform's long-term sustainability and resilience.

6.1 Rising Cybersecurity and Fraud Risks

The exponential growth in UPI transactions has also led to a rise in digital payment fraud. Criminals increasingly exploit social engineering techniques such as phishing (deceptive messages used to steal data), fake payment requests, QR code scams, and impersonation to deceive users into authorising fraudulent transactions. While UPI itself employs robust security measures, including two-factor authentication and bank-grade encryption, human error remains the weakest link.

To mitigate these risks, the Reserve Bank of India, NPCI, banks, and payment service providers continue to strengthen fraud detection systems, introduce transaction monitoring using artificial intelligence, and conduct nationwide digital awareness campaigns. However, improving financial literacy and user awareness remains a critical component of combating cyber fraud.

6.2 Infrastructure and Scalability Challenges

UPI now processes billions of transactions every month, placing significant demands on India's digital payment infrastructure. As transaction volumes continue to grow, maintaining high system availability and low processing times becomes increasingly challenging. Temporary service disruptions caused by server overloads, technical maintenance, or network issues can affect multiple payment applications simultaneously due to their reliance on shared banking infrastructure.

To address these concerns, NPCI and participating banks continue to invest in cloud infrastructure, redundant systems, and improved disaster recovery mechanisms to ensure uninterrupted payment services.

6.3 Market Concentration and Competition

Although UPI is an open and interconnected platform, a significant proportion of transactions are processed through a small number of third-party applications, particularly PhonePe and Google Pay. This concentration raises concerns regarding market competition, systemic dependence, and consumer choice.

To promote a more balanced ecosystem, regulators have explored measures such as transaction volume caps and encouraged greater participation from banks and emerging fintech companies. Sustaining healthy competition will remain important to fostering innovation and reducing systemic risks associated with excessive market concentration.

6.4 Sustainability and Revenue Generation

One of UPI's defining strengths is that most transactions are free for consumers and merchants. While this has accelerated adoption, it also presents challenges regarding the long-term financial sustainability of payment service providers. Unlike card-based payment systems, UPI generates limited direct revenue, compelling many fintech companies to rely on adjacent services such as lending, insurance, wealth management, and merchant solutions for profitability.

As transaction volumes continue to increase, developing sustainable business models without compromising affordability will be an important consideration for both industry participants and policymakers.

6.5 Privacy and Data Governance

The increasing digitisation of financial transactions has resulted in the generation of vast amounts of payment data. While this data enables innovation in credit assessment, financial analytics, and personalised services, it also raises concerns regarding data privacy, consent, and responsible usage.

India's evolving data protection framework seeks to ensure that payment data is collected, stored, and processed securely while safeguarding user rights. Maintaining public trust will require strong cybersecurity standards, transparent data governance practices, and compliance with emerging privacy regulations.

Chapter 7: Global Expansion of UPI

7.1 UPI Beyond India's Borders

Over the past decade, UPI has evolved from a domestic real-time payment system into an internationally recognised digital payments platform. As India's digital payment ecosystem matured, the National Payments Corporation of India (NPCI), through its international arm NPCI International Payments Limited (NIPL), began collaborating with foreign governments, financial institutions, and payment networks to extend UPI's reach beyond India.

Initially, the objective was to simplify payments for Indian travellers and members of the Indian diaspora. Over time, these partnerships expanded to support cross-border remittances, merchant payments, and interconnectivity between national payment systems. This reflects India's broader vision of positioning Digital Public Infrastructure (DPI) as a model for global financial inclusion.

7.2 International Partnerships and Cross-Border Payments

UPI is now operational, either fully or partially, in several countries through bilateral partnerships and payment linkages. Markets such as Singapore, the United Arab Emirates, France, Bhutan, Nepal, Sri Lanka, Mauritius, and Trinidad & Tobago have introduced UPI-based payment acceptance or are integrating with the platform. These collaborations enable Indian users to make payments abroad using familiar UPI applications while also facilitating faster and more affordable cross-border transactions.

One of the most significant milestones was the linkage between UPI and Singapore's PayNow, launched in 2023. This integration allows users in both countries to transfer money directly between bank accounts using mobile numbers or virtual payment addresses, reducing dependence on traditional remittance channels and lowering transaction costs.

Such partnerships demonstrate how connected payment systems can improve financial connectivity while supporting tourism, trade, and international remittances.

7.3 UPI as a Digital Public Infrastructure Export

India increasingly views UPI as not merely a payment application but as a component of its Digital Public Infrastructure (DPI) that can be adapted by other countries. Several nations have expressed interest in adopting or learning from India's approach to instant payments, attracted by its scalability, connectivity, and relatively low implementation costs.

Unlike proprietary payment networks, UPI operates as an open, bank-led ecosystem that encourages competition while maintaining interoperability. This governance model has

attracted attention from policymakers seeking affordable and inclusive payment infrastructure, particularly in emerging economies.

By sharing technical expertise and collaborating through NIPL(NPCI International Payments Limited), India is strengthening its position as a provider of digital public infrastructure and expanding its role in international digital cooperation.

7.4 Economic and Strategic Significance

The international expansion of UPI carries both economic and strategic benefits. For Indian travellers and businesses. Cross-border UPI acceptance simplifies overseas payments, reduces foreign exchange friction, and lowers remittance costs. These efficiencies are particularly valuable given that India remains one of the world's largest recipients of inward remittances.

From a geopolitical perspective, promoting UPI internationally enhances India's influence in shaping global digital payment standards. It also supports broader initiatives aimed at strengthening economic ties with partner countries through digital connectivity and financial cooperation.

While Visa, Mastercard, and other established payment networks continue to dominate international card payments, UPI offers an alternative model centred on interconnectivity, lower transaction costs, and public digital infrastructure rather than proprietary networks.

7.5 Challenges to International Expansion

Despite its growing international presence, UPI's global expansion faces several challenges. Differences in financial regulations, payment standards, cybersecurity requirements, and data governance frameworks require careful coordination between participating countries. Adoption also depends on local banking infrastructure, merchant acceptance, and consumer awareness.

In addition, integrating UPI with diverse payment ecosystems requires sustained technical collaboration and regulatory alignment. Addressing these challenges will be essential for ensuring seamless and secure cross-border payment experiences.

Chapter 8: The Future Outlook of UPI

8.1 From a Payment Platform to Financial Infrastructure

Over the past decade, UPI has evolved from a real-time payment system into the foundation of India's digital financial ecosystem. Looking ahead, its role is expected to expand beyond facilitating payments to supporting a broader range of financial services, including credit, embedded finance, digital identity verification, and cross-border transactions. As India's digital economy continues to grow, UPI is likely to serve as the underlying infrastructure connecting banks, fintech firms, businesses, government services, and consumers.

This transition reflects a broader shift in the payments industry, where digital payment platforms are increasingly becoming integrated financial ecosystems rather than standalone transaction networks. The next phase of growth will focus on creating a secure, inclusive, and globally connected payments ecosystem supported by technological innovation and robust regulatory frameworks.

8.2 Innovation Through Emerging Technologies

Technological innovation will play a central role in shaping the next generation of UPI-enabled services. Agentic AI is one of the most significant developments expected to transform payment operations. Unlike traditional AI systems that primarily analyse data, agentic AI can make autonomous decisions aligned with predefined objectives, enabling faster fraud detection, automated compliance, intelligent customer support, and more efficient payment processing.

Alongside AI, technologies such as blockchain and distributed ledger systems are expected to improve transaction security, accelerate settlements, and simplify identity verification. These innovations have the potential to reduce operational costs while enhancing the speed and reliability of digital financial services.

8.3 Expansion of Cross-Border and Digital Currency Ecosystems

The future of UPI will increasingly be shaped by its international expansion. Building on existing partnerships with countries such as Singapore, the UAE, France, Nepal, and Bhutan, NIPL (NPCI International Payments Limited) is expected to deepen cross-border payment connectivity. These collaborations aim to reduce remittance costs, simplify international merchant payments, and improve financial integration for travellers and the Indian diaspora.

The growing importance of India's Central Bank Digital Currency (CBDC), or Digital Rupee. As integration between UPI and CBDC improves, programmable payments, offline transactions, and cross-border settlements could become more efficient. While CBDC

adoption remains at an early stage, continued pilot programmes and regulatory support indicate its growing role within India's digital payment infrastructure.

8.4 Towards Inclusive and Embedded Finance

Future developments will focus not only on expanding digital payments but also on making them more accessible and being seamlessly integrated into everyday economic activities. Payment services are increasingly becoming embedded within e-commerce platforms, mobility applications, healthcare services, and other digital ecosystems, allowing users to complete financial transactions without leaving the primary service platform.

At the same time, regulators and industry participants are placing greater emphasis on accessibility. Payment systems are being redesigned to better serve people with disabilities through features such as enhanced authentication methods and more inclusive user interfaces. These developments align with the broader objective of ensuring that digital financial services remain accessible to every segment of society.

8.5 Building a Secure and Sustainable Payments Ecosystem

Despite rapid technological progress, the long-term success of UPI will depend on maintaining user trust and system resilience. Cybersecurity, fraud prevention, regulatory compliance, and customer experience as the most critical success factors for the future of digital payments. As transaction volumes continue to grow, investment in AI-driven risk management, stronger data governance frameworks, and resilient payment infrastructure will become increasingly important.

Furthermore, expanding digital infrastructure in underserved regions, developing new UPI use cases, and supporting innovation in lending and embedded finance will ensure that digital payments continue to contribute to India's broader economic development.

Chapter 9: Conclusion

Over the past decade, the Unified Payments Interface (UPI) has completely changed the way people make payments in India. What started in 2016 as a simple platform for instant bank transfers has grown into one of the world's largest real-time digital payment systems. By allowing people to transfer money instantly between different banks through a single platform, UPI has made digital payments faster, simpler, and more accessible for millions of users.

The success of UPI goes beyond the number of transactions it processes. It has helped bring more people into the formal banking system, supported small businesses by making digital payments affordable, and encouraged the rapid growth of India's fintech industry. It has also made everyday payments, whether for shopping, utility bills, public transport, or government services, more convenient and efficient. Together with initiatives such as Aadhaar and DigiLocker, UPI has become an important part of India's Digital Public Infrastructure.

At the same time, UPI's rapid growth has created new challenges. As more people rely on digital payments, protecting users from fraud, strengthening cybersecurity, maintaining reliable payment systems, and ensuring responsible use of customer data have become increasingly important. Policymakers and payment providers must continue to address these issues while encouraging innovation and healthy competition within the digital payments market.

UPI's success has also attracted global attention. Through partnerships with several countries, India is extending UPI beyond its borders to make cross-border payments faster and more affordable. These developments not only benefit Indian travellers and businesses but also demonstrate how India's approach to digital payments can serve as a model for other countries seeking inclusive and efficient payment systems.

Looking ahead, UPI is expected to play an even bigger role in the digital economy. New developments such as Agentic AI, embedded finance, cross-border payment networks, and the Digital Rupee are likely to expand its capabilities beyond simple money transfers. However, the future success of UPI will depend on maintaining public trust through strong security, effective regulations, and continuous technological innovation.

In conclusion, the first ten years of UPI represent much more than the growth of a successful payment platform. They demonstrate how technology, supported by effective public policy and collaboration between banks, regulators, and private companies, can improve financial access, encourage innovation, and support economic growth. As India continues its digital transformation, UPI is well positioned to remain at the centre of the country's financial system while inspiring the development of similar payment systems around the world.